

§10.1 Curves Defined by Parametric Equations

Homework : 12,24,28,39

Parameter Equations : $x = f(t)$, $y = g(t)$.

Parametric Curve : $\{ (x, y) : x = f(t) , y = g(t) \}$.

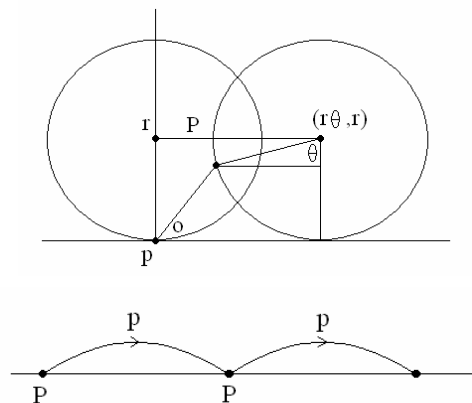
Example 1 : $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$

Solution :

Curve: the circle $x^2 + y^2 = 1$.

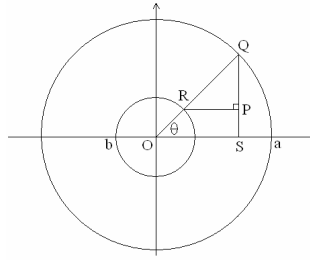
Example 2 : The curve traced out by a point P on the circumference of a circle as the circle rolls along a straight line is called a cycloid. Find parametric equations for the cycloid.

Solution :



$$\begin{aligned} p(x, y) \\ x &= r\theta - r \sin \theta \\ y &= r - r \cos \theta. \end{aligned}$$

Example 3 : Find $P(x, y)$ in terms of Q . What is its parametric curve?



Solution :

$$x = \overline{SO} = a \cos \theta, \quad y = \overline{QS} - \overline{SP} = a \sin \theta - b \sin \theta$$

$$\Rightarrow \cos \theta = \frac{x}{a} \quad \text{and} \quad \sin \theta = \frac{y}{(a-b)}$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{(a-b)^2} = 1.$$